

Adnan Ahmed Khan

Senior Backend & Platform Engineer — Distributed Systems, Kubernetes, SRE

Islamabad, Pakistan | Open to Remote & Employer-Sponsored Relocation

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WORK EXPERIENCE

HackOnTech

Senior Software Engineer (Backend & Platform)

2023 – Present

Internal B2B/G2G platforms for international NGOs; Scale: 50k–100k DAU, 500–800 RPS peak, 99.9% availability target.

- Decomposed a monolith into 6 Docker/EKS microservices, cutting deploy time from hours to minutes with zero-downtime releases.
- Instrumented Jaeger tracing and Prometheus/Grafana dashboards to cut MTTR from ~45 minutes to under 10 by isolating cross-service latency.
- Implemented Temporal workflows with idempotency/compensation to cut failed cross-service ops ~80% (150/day to <30).
- Reduced payment failures from ~5% to 2% by rebuilding retries and rollbacks in AWS Step Functions.
- Reduced core query latency from 30s+ to under 1s through indexing, connection pooling, and service decomposition (Postgres/Redis).
- Built GitHub Actions + ArgoCD CI/CD with security gates and canary/blue-green releases, supporting 99.9% availability.
- Reduced AWS spend ~25% via EKS node rightsizing, autoscaling threshold tuning, and spot for non-prod without SLO impact.
- Stabilized 500–800 RPS burst windows with rate limiting, queue-based smoothing, and graceful degradation.

Tech: AWS (EKS, Lambda, S3, Secrets Manager, IAM, CloudFront, Route 53), IaC (Terraform, Helm), CI/CD (GitHub Actions, ArgoCD), Observability (Prometheus, Grafana, Jaeger, Datadog), Backend (TypeScript/Node.js, Nest.js, Python/FastAPI), Workflows (Temporal, Step Functions), Datastores (Postgres, Redis), Containers (Docker).

Creative IT Park

Backend Developer

2021 – 2023

- Delivered backend services for School Management System, Rawalpindi Women University, and Capital Development Authority portal, building APIs and database workflows.
- Cut search latency by eliminating N+1 queries and introducing aggregated queries with Elasticsearch + Redis caching.
- Maintained notification delivery during downstream failures by implementing a Kafka-based pipeline with retries and circuit breakers.
- Prevented traffic to degraded pods by operating Kubernetes/Helm deployments with readiness probes and health gating.
- Secured healthcare data pipelines by implementing AWS Glue ETL with encryption at rest/in transit and fine-grained IAM controls.

Tech: Backend (Node.js, Python/FastAPI), Frontend (React, TypeScript), AWS (ECS, RDS, DynamoDB, Glue, IAM), Data/Queues (Kafka, Elasticsearch, Redis), Kubernetes/Helm, Observability (ELK Stack), Datastores (PostgreSQL, MongoDB), Testing (Jest), Containers (Docker).

GeekGenix

Backend Developer

2020 – 2021

- Designed payment APIs with Express.js/MongoDB and integrated Stripe checkout plus webhook reconciliation for production use.
- Hardened authentication with Passport.js strategies and Jest tests, stabilizing login flows in QA.

Tech: Node.js, Express.js, MongoDB, React, Passport.js, Stripe API, Jest.

EDUCATION

BS Computer Science

PMAS Arid Agriculture University, Rawalpindi

SELECTED PROJECTS

openclaw-workflow

 | Open Source

- Built a TypeScript/Node.js OpenClaw plugin for YAML/JSON-driven workflow orchestration with dependency management, parallel execution, retries, output gates, and partial resume.
- Implemented an authoring-to-execution compiler, workflow executor, sealed step runners, artifact/state backends, Redis support, and Node test/build pipeline.

Vidx

 | 2025

- Built a 4K-capable rendering platform with a FastAPI backend, async task queues, distributed GPU workers, FFmpeg/GL-Transitions, and 50+ shaders.
- Compiled a custom FFmpeg build and streaming pipelines to cut peak memory ~40%, enabling 4K on mid-tier GPUs under burst load.

Crackq.me

 | 2023

Authorized Security Research & Educational Platform.

- Built a distributed hash-testing platform for authorized security research with Hashcat, secure job dispatch, a BOINC-style volunteer GPU grid, and capability-aware routing.
- Tuned kernel batch sizes, scheduling, and priority queues to increase hashes/sec without adding hardware while keeping urgent jobs moving.

pdf-annotate.js

 | 2024

- Built WebSocket real-time collaboration for PDF annotations with conflict resolution, synchronized sessions, StoreAdapter changes, spatial indexing, and SVG optimizations for 60fps rendering.